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A Monte Carlo study on LCCC characterization of graft copolymers at the critical condition of side chains

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ABSTRACT

Liquid chromatography at the critical condition (LCCC) has been used to characterize graft copolymers based on the assumption that one of the copolymer blocks becomes chromatographically "invisible" at the critical condition of the corresponding homopolymer. We investigate the validity of this assumption with lattice Monte Carlo simulations of A-g-B graft copolymers modeled as either random walk (RW) or self-avoiding walk (SAW) chains composed of multiple invisible B blocks grafted to an A backbone that is either in size-exclusion chromatography (SEC) or liquid adsorption chromatography (LAC) mode. The simulations show that, in agreement with recent experimental results, the B blocks have a small, but noticeable, impact on the overall elution of the graft copolymer. This influence exists even when the chains are modeled as RWs, indicating that its fundamental origin is due to chain connectivity, not excluded volume interactions. In general, both models show that grafting B blocks to an A backbone in SEC mode tends to increase copolymer retention, while grafting B blocks to an A backbone in LAC mode tends to decrease copolymer retention. When the copolymer is modeled as a SAW, the excluded volume interaction increases the entropic penalty associated with the addition of a B block and, therefore, the addition of B blocks in SAW chains is more likely to result in a decrease in the retention time of the copolymer when the A backbone is in LAC mode.

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1. Introduction

Recent advances in synthesis technologies have enabled the creation of a large number of novel complex polymers, such as block, star, dendritic, cyclic and graft polymers [1–7]. Among these, graft copolymers containing many side chains chemically grafted onto a linear backbone have received great attention because of fascinating properties such as wormlike conformations, compact molecular dimension, and significant chain end effects that arise from their confined and compact structures. However, synthesized graft polymers are usually mixtures of polymers that vary not only in molar mass, but also in chain architecture and/or chemical composition. Separation and characterization of graft copolymers to obtain samples with high purity in both chain architecture and molar mass is very important for establishing structure property relationships for subsequent applications.

At present, size-exclusion chromatography (SEC) remains the most commonly used technique to determine the molecular weight distribution of graft copolymers according to their hydrodynamic volume in solution. However, SEC cannot effectively separate graft copolymers that have both molecular weight and chain architecture distributions [8,9]. In recent years, some new liquid chromatographic methods other than SEC, such as liquid chromatography at the critical condition (LCCC) [8,10,11] and liquid adsorption chromatography (LAC) [8], have been developed for the characterization of complex polymers according to their chain architecture and/or chemical composition. LCCC has been shown to be a very effective method to characterize individual blocks in linear block copolymers [12–14] as well as graft copolymers [11,15,16]. Separation of block or graft copolymers with LCCC is based on the assumption that one of the blocks in the copolymer can be made chromatographically "invisible" at the critical condition of the corresponding homopolymer; thus, the retention of the

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copolymer will be determined solely by length of the "visible" block [17–19]. However, some chromatographic separations of block [20] or graft copolymers [21,22] have begun to challenge this assumption. For example, Chang and coworkers [21] investigated the separation of deuterated PS-graft-hydrogenous PS (hPS-g-dPS) copolymers having a single hPS backbone and a few dPS side chains. In that work, the graft copolymer is separated under conditions in which dPS branches are at the critical condition and are purportedly chromatographically invisible, while the visible hPS backbone interacts more strongly with the stationary phase and is in LAC mode. They observed an anomalous phenomenon: the retention of the graft copolymer decreased as the number of "invisible" dPS branches increased. More recently, Chang and coworkers [22] examined the retention behavior of a hPS-b-dPS block copolymer under similar conditions and observed the same interesting retention behavior, namely, a decrease in retention time as the invisible block length increases when the overall retention of the block copolymer is in LAC mode.

Typically, the retention time of a polymer decreases as its size or molecular weight increases in SEC mode, while retention time increases as its size increases in LAC mode. However, in Chang's experiment with graft copolymers, the polymer displays features of both modes: the retention time of the graft copolymer decreases as the number of branches and overall molecular weight of the polymer increases (i.e., the polymer behaves as if it is in SEC mode) even though the graft copolymer elutes after the solvent peak, a characteristic of LAC mode. One potential explanation of this phenomenon is the excluded volume effect. As the number of invisible branches increases, it is more difficult for hPS-g-dPS to enter pores because of the excluded volume effect and hence the retention time of the polymer deceases. Another potential explanation is effects due the chain connectivity, as suggested in our recent joint computational and experimental study of hPS-b-dPS block copolymers [22]. In the current study, we apply Monte Carlo (MC) simulations to examine the LCCC separation of A-g-B copolymers while setting the A backbone in the visible mode and the B side chain at the chromatographically invisible mode. We use two chain models, Random Walk (RW) and Self-Avoiding Walk (SAW) models, to describe the graft copolymers. We examine the relative importance of the contributions of chain connectivity and the excluded volume interaction to the interesting retention behavior that has been observed for graft copolymers.

2. Monte Carlo simulation methods

In the current study, both random walk (RW) and self-avoiding walk (SAW) chains are used to model graft copolymers in a simple cubic lattice with a coordination number $z = 6$. In the RW model, different polymer beads can occupy the same site on the lattice, while no two polymer beads may occupy the same site in the SAW model. Unless otherwise stated, simulations of an A-g-B copolymer composed of an A backbone of length, N_A, 50 and B side chains of length, N_B, 22 are performed in a slit pore with dimensions of $250a \times 250a \times 30a$, where a is the lattice unit length. Periodic boundary conditions are applied in the X and Y directions and there are two impenetrable walls in the $Z = a$ and $Z = 30a$ planes representing the slit pore surface. In some simulations, L_Z, the distance between the walls along the Z direction, is varied to explore the impact of changing the pore size on the elution behavior. Typical parameters used in the simulations are summarized in Table 1.

The biased chain insertion method is used to determine the standard chemical potential of a graft copolymer in the slit pore. First, we randomly select one site in the slit pore (between the $Z = 2a$ and $Z = 29a$ planes inclusive) to be the location of the first bead of the graft copolymer backbone. The remaining beads of both the backbone, which is composed of A-type beads, and the grafts, which are composed of B-type beads, are then grown using the biased chain insertion method. Chain beads in the $Z = 2a$ and $Z = 29a$ planes of the simulation box are considered to be adsorbed on pore surfaces. For all adsorbed beads, reduced polymer-surface interaction energies, $\varepsilon_W(A)$ and $\varepsilon_W(B)$, are applied for A and B monomers, respectively. In this work, the A backbone are made the "visible" block by setting $\varepsilon_W(A)$ to appropriate values to represent either SEC or LAC modes. The B side chains are made "invisible" by setting $\varepsilon_W(B)$ to the critical condition of a homopolymer. Our previous simulation studies [23–25] have established that the polymer surface interaction energy at the critical condition, ε_W^{cc}, is −0.18232 for an RW chain and −0.276 for a SAW chain, respectively.

Table 1
Typical simulation parameters.[a]

N_A	N_B	L_Z	ε_W^{cc}(RW)	ε_W^{cc}(SAW)
50	22	$30a$	−0.18232	−0.276

[a] See text for meaning of symbols.

The standard chemical potential of the graft copolymer chain in the slit pore, μ_{in}^0, is determined from the Rosenbluth weighting factor after the chain is inserted in the pore. Detailed information for the calculation of the standard chemical potential can be found in previous simulation studies [24–32]. The partition coefficient K is then calculated by the equation $\ln K = -(\mu_{in}^0 - \mu_{bulk}^0)$, where μ_{bulk}^0 is the standard chemical potential of the chain in the bulk solution that was determined from simulations in a cubic lattice of dimension $100a \times 100a \times 100a$ with periodic boundary conditions applied in all three directions. In order to investigate how the B branches impact copolymer retention, we also examined $K(B^*)$, the partition coefficient of only the B side chain of an A-g-B copolymer, for both the RW and SAW models. To determine $K(B^*)$, an A-g-B copolymer composed of an A backbone and a single B side chain was grown in the slit pore and bulk simulation boxes using the typical method, and the chemical potential of the B side chain was determined from the Rosenbluth factor using only the B beads of the chain. $K(B^*)$ was then calculated using the equation relating the partition coefficient and chemical potential given above. In particular, we were interested in how $K(B^*)$ depended on the location of the graft point in the pore. To investigate this dependence, we determined the Z layer of the graft point of each B side chain after the chain was grown in the slit and, then, calculated $K(B^*)$ for a given Z layer by averaging over all chain insertions where the graft point was located in the appropriate layer.

3. Result and discussion

3.1. RW chain model

All the results presented in this section are obtained with the RW chain model. We first consider an A-g-B copolymer consisting of a single A-backbone and several B branches uniformly grafted onto the A backbone. In the current study, the B branches are the invisible block, while the A backbone is the visible block, which is either in SEC mode or LAC mode. To match Chang's experiments [21], we set the length of A backbone, $N_A = 50$ and the length of B side chains, $N_B = 22$, respectively, and vary the number of side chains from 0 to 5. Fig. 1 presents the dependence of K(A-g-B), the partition coefficient of an A-g-B graft copolymer, on the number of invisible B branches. The plot includes a data point with zero B branches, leaving only the A backbone. The partition coefficient K(A-g-B) determines the elution volume of the graft copolymer in

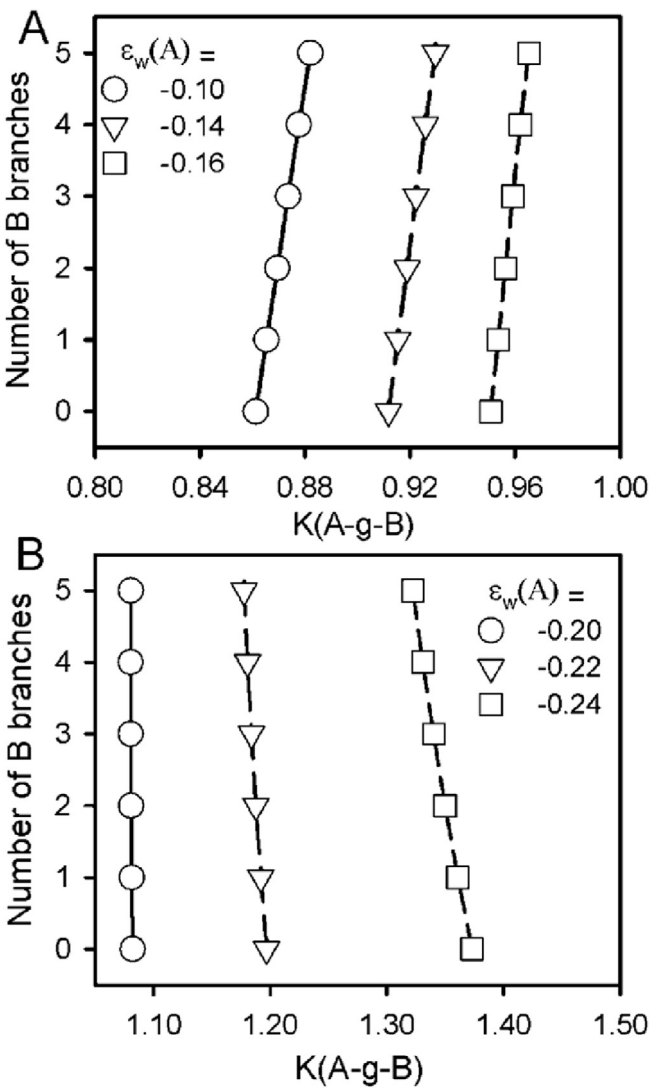

Fig. 1. The number of B branches versus K(A-g-B) when the B branches are in the critical condition ($\varepsilon_w(B) = -0.1832$) and the A backbone is (a) in SEC mode and (b) in LAC mode. The chains are random walks composed of an A backbone of length, $N_A = 50$ and B side chains of length, $N_B = 22$. The legend indicates values of ε_w(A).

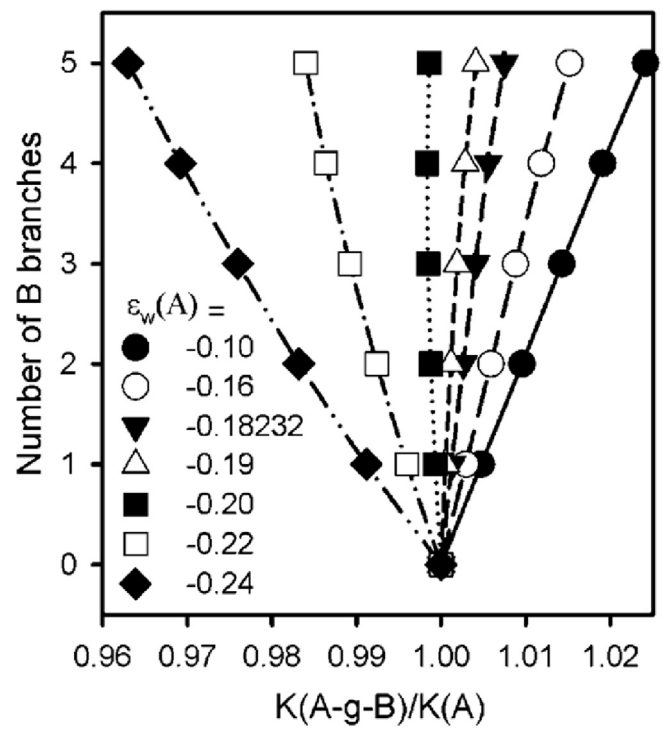

Fig. 2. The number of B branches vs. K(A-g-B)/K(A) while varying the A backbone from SEC mode to LAC mode, with values of ε_w(A) indicated by the figure legend. The B branches are at the critical condition ($\varepsilon_w(B) = -0.18232$). The chains are random walks composed of an A backbone of length, $N_A = 50$ and B side chains of length, $N_B = 22$.

chromatography experiments. If K(A-g-B) = 1, the graft copolymer would appear to elute at the solvent peak. Typically, K(A-g-B) < 1 corresponds to SEC mode and K(A-g-B) > 1 corresponds to LAC mode. When the A backbone is in SEC mode (Fig. 1a), we observe an increase in K(A-g-B) with the increase in the number of B branches. In contrast, when the A backbone is in LAC mode (Fig. 1b), the retention of the copolymer, K(A-g-B), decreases as the number of B branches increases. The latter behavior (a decrease in retention time as the overall molecular weight of the copolymer increases when the backbone is in LAC mode) is the same phenomenon that was recently seen in Chang's experiment on LCCC separation of hPS-g-dPS comb-shaped copolymers [21].

To more clearly show the impact of B branches on the elution of the copolymer, Fig. 2 presents the number of B branches versus K(A-g-B)/K(A), where K(A) is the partition coefficient of an A homopolymer of length 50 for several values of ε_w(A). For reference, K(A), the partition coefficient of the A backbone alone, for the values of ε_w(A) included in Fig. 2 are provided in Table 2. If K(A-g-B)/K(A) is not exactly one, the invisible B branches alter the retention of the graft copolymer. As discussed previously, when the A backbone is in SEC mode, the addition of B grafts increases K(A-g-B) and K(A-g-B)/K(A) > 1, while the addition of B grafts decreases K(A-g-B) and K(A-g-B)/K(A) < 1 when the A backbone is in LAC mode. The transition from invisible B branches increasing K(A-g-B) to decreasing K(A-g-B) does not occur exactly at the critical condition (when $\varepsilon_w(A) = \varepsilon_w^{cc} = -0.18232$), but instead occurs for ε_w(A) between −0.19 and −0.2, when the A backbone is slightly in the LAC mode. Thus, K(A-g-B) increases as B side chains are added when all of the beads of the chain are at the critical condition of a homopolymer ($\varepsilon_w(A) = \varepsilon_w(B) = \varepsilon_w^{cc} = -0.18232$). Additionally, in both SEC and LAC modes, the impact of the B branches on elution of the copolymer is stronger when ε_w(A) is far from ε_w^{cc}.

In the past, it was generally believed that, within the RW model which ignores the excluded volume interaction, grafts at the critical condition do not impact the retention of the backbone, and the retention time of a copolymer will be the same as the retention time of the backbone only, i.e., K(A-g-B) = K(A). From the above data, it is clear that this is not the case and that the influence of the purportedly invisible B branch on graft copolymer retention exists even for RW chains with no excluded volume interaction. Furthermore, the results shown in Figs. 1 and 2 indicate that the addition of invisible B grafts (i.e., B grafts are at the critical condition) has the potential to either increase or decrease K(A-g-B) depending on the elution mode of the A backbone: K(A-g-B) is typically greater than K(A) when the A backbone is in SEC mode and less than K(A) when the A backbone is in LAC mode. We attribute this different behavior to the B block's influence on elution being

Table 2
K(A) for various values of ε_w(A) for Random Walk chains.

ε_w(A)	−0.10	−0.16	−0.18232	−0.19	−0.20	−0.22	−0.24
K(A)	0.861	0.951	1.012	1.040	1.082	1.197	1.373

dependent on the location of the graft point in the pore. Specifically, if an invisible B side chain is attached to the A backbone through a graft point that is in contact with a pore surface, the presence of the graft will increase the repulsion of the chain from the pore, while B side chains attached to graft points that are not in direct contact with pore surfaces will increase attraction of the chain to the pore. Such behavior can be quantified by determining the free energy of the B branches in the pore as a function of the graft point position in the Z layer. Specifically, we determined $K(B^*)$, the partition coefficient of only the B side chain of an A-g-B copolymer, as a function of the location of the graft point (Fig. 3). $K(B^*)$ in the RW model depends on graft point position in the following ways:

(1) If the side chain is attached to a graft point that is adsorbed on a pore surface (i.e., in the $Z = 2$ or $Z = 29$ layers), $K(B^*) < 1$. Grafting a side chain to an adsorbed graft point reduces chain entropy even when the copolymer is modeled as an RW chain without excluded volume interactions. This entropic penalty results from the location of the first bead of the B branch (i.e., the bead connected to the graft point) being restricted. As the graft point is in contact with the surface, one of its nearest neighbors is occupied by a surface site and the first bead of the B branch is forbidden from occupying this site. Thus, there is an entropic penalty for grafting a branch at the surface even in RW chains, causing $K(B^*)$ in the $Z = 2$ and $Z = 29$ layers to be much less than 1 in Fig. 3. B branches attached to adsorbed graft points will, therefore, decrease the attraction between the copolymer and the pore, tending to result in K(A-g-B) < K(A).

(2) When the graft point is near but not directly adjacent to the pore surface (i.e., the graft point is not in the $Z = 2$ or $Z = 29$ layers), $K(B^*) > 1$. As we [33] and others [34] have shown, the critical condition for RW chains of finite length does not occur when $K = 1$, as predicted for infinitely long Gaussian chains, but instead occurs when K is slightly larger than 1. This effect is due to the free energy of the end beads of a chain at the critical condition being slightly lower than the free energy of the middle beads. Thus, even though the B branches in the A-g-B copolymer are at the critical condition, the free energy of the B branches will be negative near the pore surface, resulting in $K(B^*) > 1$ and hence K(A-g-B) > K(A). Also, as the number of branches increases (with an accompanying increase in the number of end beads), this effect becomes more significant and K(A-g-B) will continue to increase [33]. $K(B^*)$ decreases and, eventually, approaches 1 as the graft point moves farther away from the surface, as it becomes less likely that the relatively short B branches will interact with the surface.

Given the features of $K(B^*)$ shown above, we can now understand the impact of B invisible branches on the K(A-g-B) when A is in both SEC and LAC modes by noting K(A-g-B) = K(A)$K(B^*)$ for the RW model. We first calculate the density profile of the graft points across the slit for several values of ε_w(A) (Fig. 4). When the A backbone is in SEC mode ($\varepsilon_w(A) = -0.10$ in Fig. 4), the density of graft points near the surface is lower than in the middle of the slit. As graft points are more likely to be located away from the surface, the majority of copolymer conformations would have $K(B^*) > 1$. The increased attraction between the copolymer chain and the pore arising from the non-surface graft points is the dominant effect, resulting in K(A-g-B) > K(A). When the A backbone is in the LAC mode, ($\varepsilon_w(A) = -0.24$), the density of the graft points is higher at the surface than in the middle of the slit. The repulsion of the chain from the pore arising from the graft points at the surface becomes the dominant effect and hence K(A-g-B) < K(A). The transition from having attraction being the dominant effect to repulsion being the dominant effect occurs near the critical condition. For example, as shown in Fig. 4, the density profile of graft points when $\varepsilon_w(A) = -0.19$ is relatively flat across the pore, and K(A-g-B)/K(A) is closer to 1.

Earlier, in collaboration with Chang's group, we examined the retention of block copolymers when one of the blocks is in the critical condition mode and the other block is visible either in SEC or LAC mode [22]. Architecturally speaking, block copolymers can be viewed as graft copolymers with a single branch that is grafted to the end of the backbone. Fig. 5 compares the influence of invisible B branches on the elution of a copolymer when the grafting position and the graft length are varied. This comparison reveals two interesting features: First, attaching a B block to the

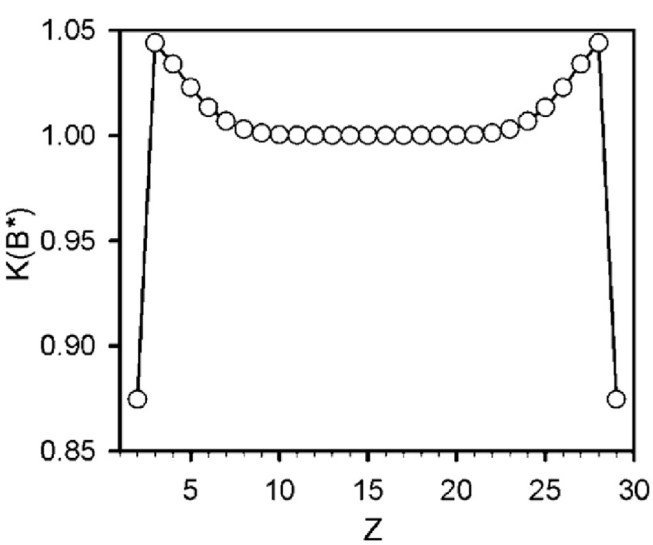

Fig. 3. $K(B^*)$, the partition coefficient of only the beads of the B branch of an RW A-g-B copolymer, as a function of the position of the graft point in the Z direction. Solid walls comprising the surfaces of the slit pore are located in the $Z = 1$ and $Z = 30$ planes. $K(B^*)$ was calculated using a single B side chain with length 22 and $\varepsilon_w(B) = -0.18232$ (the critical condition) grafted to an A backbone of length 50.

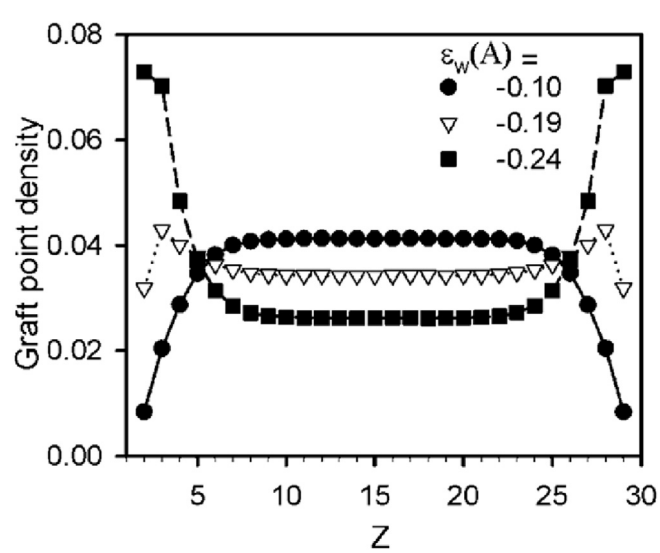

Fig. 4. Density profiles of the graft point of an RW A-g-B copolymer for the values of ε_w(A) indicated in the figure legend. The copolymer chains have $N_A = 50$ and $N_B = 22$ with $\varepsilon_w(B) = -0.18232$.

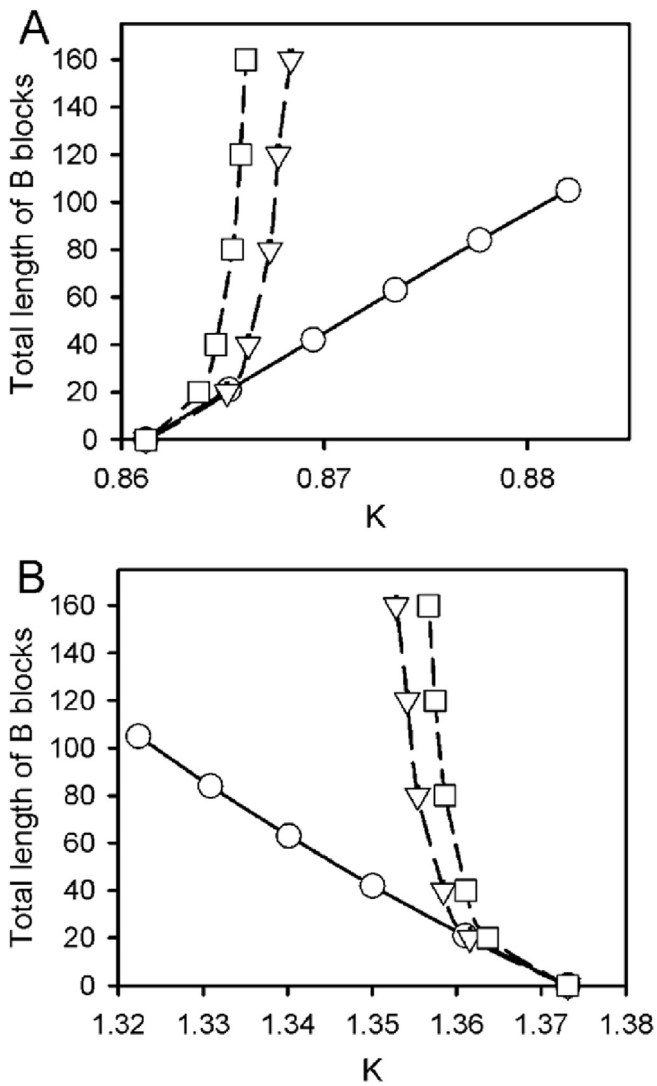

Fig. 5. Impact of grafting location and graft length on K of copolymers composed of B blocks at the critical condition attached to an A backbone in SEC mode (panel a where $\varepsilon_w(A) = -0.10$) and LAC mode (panel b where $\varepsilon_w(A) = -0.24$). Square symbols correspond to a single B block with length indicated in the figure attached to the end of the A backbone, creating a block copolymer. Data for graft copolymers are shown with triangles and circles for the cases of a single B side chain grafted to the middle of the A backbone and multiple B side chains of length 22 uniformly grafted along the A backbone, respectively.

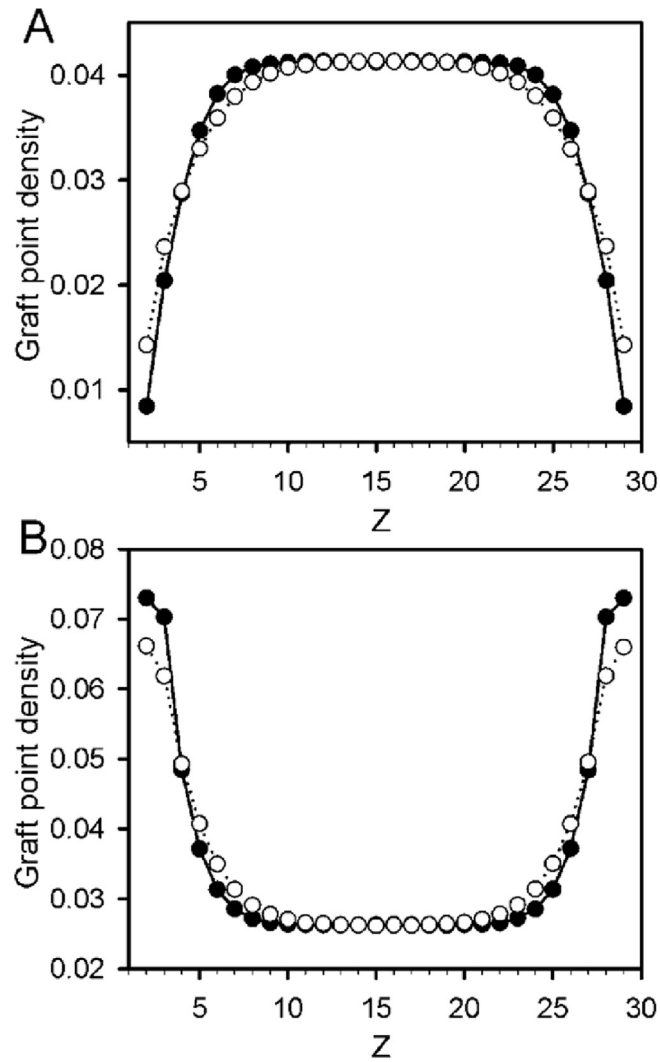

Fig. 6. Density profiles of the graft point of RW A-g-B (closed symbols) and the junction point of AB block copolymers (open symbols) for $\varepsilon_w(A) = -0.10$ (panel a) and -0.24 (panel b). The copolymer chains are composed of an A backbone of length 50 and a single B branch of length 22 that is either attached to the middle (A-g-B copolymer) or end (AB block copolymer) of the A backbone. The B block is at the critical condition ($\varepsilon_w(B) = -0.18232$).

middle of the A backbone (i.e., creating a graft copolymer) has a stronger impact on the retention of the copolymer than attaching a B block at the end of the A backbone (i.e., creating a block copolymer), given the same grafting length. Second, grafting several short B branches has a stronger impact on retention than attaching a single long B branch when the total number of B beads in the copolymer is held constant. Both features are present when the A backbone is in SEC mode (Fig. 5a) and when the A backbone is in LAC mode (Fig. 5b).

The first feature can be understood in terms of the variation in $K(B^*)$ as a function of graft position. The density profiles of graft points at the middle of the backbone (i.e., as in A-g-B copolymers) are compared with those located at the end of the backbone (i.e., block copolymers) in Fig. 6. When the A backbone is in SEC mode ($\varepsilon_w(A) = -0.10$, Fig. 6a), the density of graft points at the surface is lower for graft copolymers than for block copolymers. Therefore, the attraction due to non-surface graft points is able to dominate the repulsion due to graft points at surfaces to a greater extent in the A-g-B copolymer than in the block copolymer and the impact of the B branch on K is greater for the A-g-B copolymer than the block copolymer. When the A backbone is in LAC mode ($\varepsilon_w(A) = -0.24$, Fig. 6b), the density of graft points at the surface is greater for graft copolymers than for block copolymers. Therefore, the repulsion due to graft points at surfaces is more significant for graft copolymers than for block copolymers and the impact of the B branch on K is greater for the A-g-B copolymer than the block copolymer. The second feature (i.e., that there is a difference between grafting a single long branch versus grafting several short branches) parallels a result from previous Monte Carlo simulations of the partitioning of star polymers into pores [33]. Star polymers can be viewed as a graft copolymer in which all the branches are grafted at the center of the backbone. As we have previously discussed, the partition coefficient of star polymers at the critical condition is not exactly 1, but is instead slightly greater than 1 and increases as the number of arms increases (Fig. 5 in Ref. [33]). Similarly, grafting multiple branches to the backbone continues to increase or decrease K(A-g-B) if the A backbone is in SEC mode or LAC mode, respectively. For

each branch that is added to the chain, an attractive end bead that increases K will be added, or an additional entropic penalty that decreases K will result if the graft point is at the surface.

The influence of the "invisible" B branch on the elution of the graft copolymer can only be felt if the B branch is able to interact with walls. As the slit width increases, it becomes increasingly likely that the copolymer will adopt a conformation within the slit in which the B branches do not directly interact with the slit surfaces. Therefore, it is expected that the deviation of K(A-g-B) from K(A) will be reduced as the slit width increases. Fig. 7 shows the dependence of K(A-g-B)/K(A) on the branch number while varying the slit width, L_z, from 20a to 50a. As expected, as the slit width increases, K(A-g-B)/K(A) shifts closer to the straight line where K(A-g-B)/K(A) = 1 both when the visible backbone is in the SEC mode and when it is in LAC mode.

3.2. SAW section

In this section, we use the SAW model to investigate A-g-B graft copolymers in which the B block is at the critical condition partitioning into pores. The B branches of a SAW chain can influence the partitioning of the copolymer not only because of chain connectivity, similar to RW model chains, but also because the SAW model accounts for excluded volume interactions that are present in real polymer chains. Fig. 8 shows the dependence of K(A-g-B)/K(A) on the number of invisible B branches where the length of the A backbone and the length of each B branch are set to 50 and 22, respectively. Table 3 provides values of K(A), the partition coefficient of the A backbone alone, for values of ε_w(A) used in Fig. 8. To a large extent, the results obtained by the SAW chain model are similar to those of the RW model (Fig. 2), as K(A-g-B)/K(A) increases

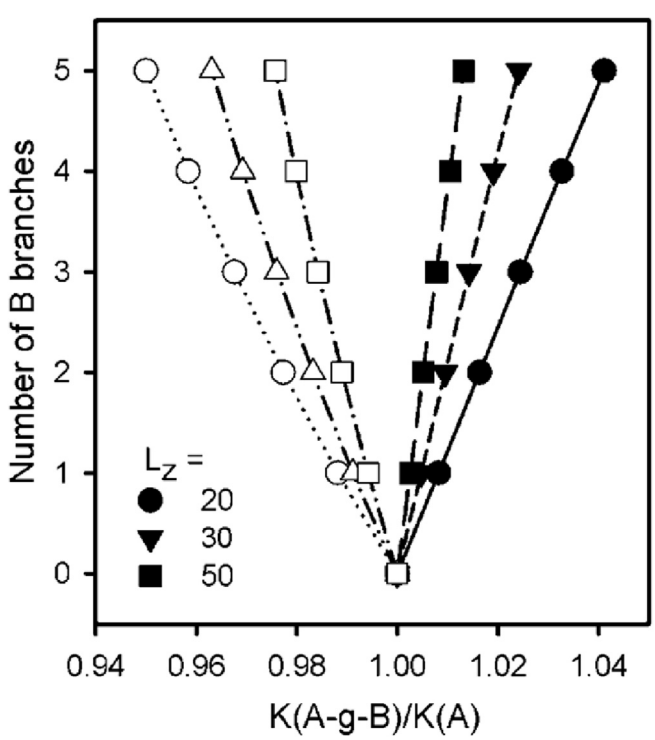

Fig. 7. The number of B branches vs. K(A-g-B)/K(A) when the width of the lattice box (L_Z) is varied from 20a to 50a, as indicated by the figure legend. The chain is an RW with N_A = 50 and N_B = 22 and B side chains at the critical condition (ε_w(B) = −0.18232). The A backbone is either in SEC mode (ε_w(A) = −0.10, shown in the right curves with closed symbols) or LAC mode (ε_w(A) = −0.24, shown in the left curves with open symbols).

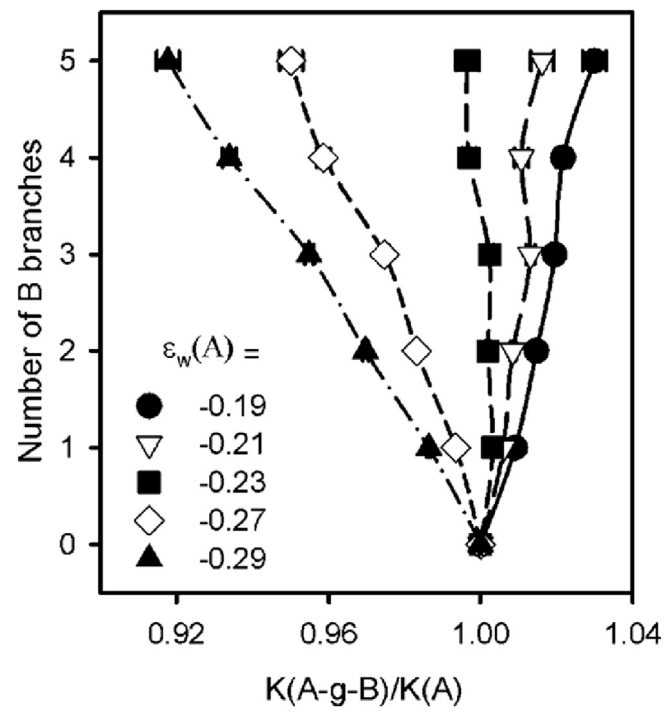

Fig. 8. The number of B branches vs. K(A-g-B)/K(A) while varying ε_w(A) from SEC mode to LAC mode, as indicated in the figure legend. ε_w(B) is at the critical condition of a B homopolymer (ε_w(B) = −0.276). The chain is a self-avoiding walk with N_A = 50 and N_B = 22.

with the number of B branches when the visible A backbone is in SEC mode (ε_w(A) = −0.19 or −0.21) and decreases when the backbone is in LAC mode (ε_w(A) = −0.29). However, Fig. 8 indicates that the addition of B side chains for SAW chains is more likely to result in repulsion of the graft copolymer from the pore. First, K(A-g-B) is as much as ~8% lower than K(A) in the SAW model when ε_w(A) = −0.29. In comparison, K(A-g-B) was only as much as ~4% lower than K(A) in the RW model, even though the RW model was used to investigate polymer-surface interactions that were more attractive relative to ε_w^{cc}. Second, the transition from B branches increasing K(A-g-B) to decreasing K(A-g-B) occurs when the A block is slightly in the SEC mode for SAW chains, while this transition was shown to occur when the A block is slightly in LAC mode for RW chains. For example, when ε_w(A) = −0.27, the A backbone of a SAW copolymer is in SEC mode, and the addition of B branches results in a decrease in K(A-g-B).

Like the RW model, the dependence of K(A-g-B) of SAW copolymers on the number of B side chains and ε_w(A) can be explained with K(B*), the partition coefficient of only the B beads of the copolymer, and the graft point density profiles. The plot of K(B*) as a function of graft point position Z for SAW chains (Fig. 9) shares much in common with the corresponding plot of the RW model, as K(B*) < 1 for graft points very near pore surfaces and K(B*) > 1 for graft points further away from the surface. However, two features of Fig. 9 indicate that there is a larger entropic penalty associated with the addition of "invisible" B side chains near pore surfaces in the SAW model. First, K(B*) for graft points at surfaces (in the Z = 2 and Z = 29 layers) is 0.73 for SAW chains, lower than the

Table 3
K(A) for various values of ε_w(A) for Self-Avoiding Walk chains.

ε_w(A)	−0.19	−0.21	−0.23	−0.27	−0.29
K(A)	0.860	0.894	0.938	1.068	1.166

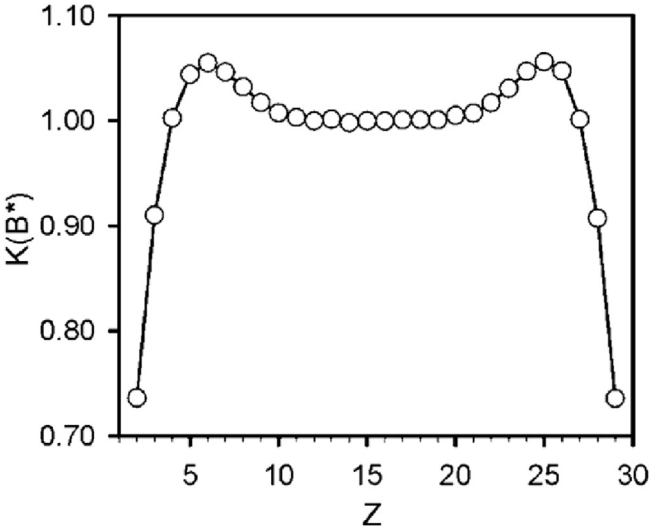

Fig. 9. $K(B^*)$, the partition coefficient of only the beads of the B branch of a SAW A-g-B copolymer, as a function of the position of the graft point in the Z direction. Solid walls comprising the surfaces of the slit pore are located in the $Z = 1$ and $Z = 30$ planes. $K(B^*)$ was calculated using a single B side chain with $N_B = 22$ and $\varepsilon_w(B) = -0.276$ (the critical condition) grafted to an A backbone with $N_A = 50$.

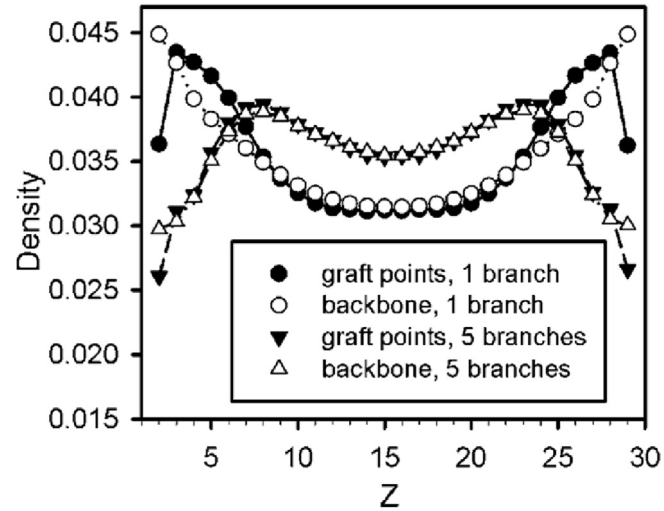

Fig. 10. Density profiles of SAW A-g-B graft copolymers with $N_A = 50$ and $N_B = 22$. The A backbone is in LAC mode with $\varepsilon_w(A) = -0.29$, while the B side chains are at the critical condition ($\varepsilon_w(B) = -0.276$). Density profiles of only the graft points or the entire A backbone for copolymers with a single B side chain or five B side chains are shown.

corresponding value of $K(B^*)$ for RW chains. Second, while the entropic effect that reduces $K(B^*)$ occurs only when the graft point is in contact with the surface for RW chains, this effect propagates several layers into the slit for SAW chains. The increased entropic penalty associated with graft points near surfaces in SAW copolymers results from the excluded volume interaction; when the graft point of a SAW chain is near a pore surface, conformations of the B side chain are restricted by both the pore surface and the beads of the copolymer. Because of the increased entropic penalty in the SAW model, the impact of the side chains at repulsive graft points (i.e., those very near surfaces where $K(B^*) < 1$) dominates that of side chains at attractive graft points (i.e., those where $K(B^*) > 1$) for less attractive values of $\varepsilon_w(A)$ than was observed using the RW model and, hence, K(A-g-B) begins to decrease with the addition of B side chains when the A backbone is still in SEC mode. For example, when $\varepsilon_w(A) = -0.23$, the A backbone is in SEC mode and the graft point density near the pore surface is low (Supplementary Fig. 1), but the impact of side chains at attractive graft points is only able to just compensate that of side chains at repulsive graft points and K(A-g-B) remains roughly constant with the addition of B side chains. If the A backbone is in SEC mode and far from the critical point ($\varepsilon_w(A) = -0.19$, Supplementary Fig. 1), the density of the graft points very near the surface becomes sufficiently low that side chains at attractive graft points are able to dominate and K(A-g-B) increases with the addition of B side chains.

Finally, we examine how the density profiles of a SAW graft copolymer when the A backbone is in LAC mode ($\varepsilon_w(A) = -0.29$) are influenced by the addition of B side chains at the critical condition (Fig. 10). When only a single B side chain is grafted to the backbone, the density of all beads of the A backbone is at a maximum at the surface, as expected for a chain in LAC mode. However, the density profile of only the graft points does not match that of the entire backbone, as the chain preferentially adopts conformations in which the addition of the side chain will be energetically favorable. Thus, the graft point density profile is, in comparison to that of the entire backbone, lower at the surface and higher a few layers away from the surface, reducing the conformations where $K(B^*) << 1$ and increasing conformations where $K(B^*) > 1$. As more side chains are grafted to the backbone, their influence increases and the backbone begins to shift away from the surfaces. When five B side chains are grafted to the backbone, the backbone density is at a minimum at the surface, despite the A backbone being in the LAC mode, reducing the number of energetically unfavorable adsorbed graft points. This result is another consequence of the excluded volume interaction, as the density profiles of RW chains do not change as dramatically as the B branches are grafted to the copolymer (Supplementary Fig. 2).

4. Conclusion

In the current study, both RW and SAW chain models were used to investigate the LCCC separation of A-g-B graft copolymers containing a "visible" A backbone and a series of "invisible" B branches. We found that the invisible B branches influence the elution of A-g-B graft copolymers when the visible A backbone is in both SEC and LAC modes. This influence exists even when the excluded volume interaction of the polymer chain is ignored and the RW chain model is used. Therefore, the simulations reveal that the basic cause of the influence of a purportedly invisible branch on the elution of a graft copolymer graft is the chain connectivity between the backbone and branches. We calculated the partition coefficient of the branch, $K(B^*)$, as a function of position of the grafting point in the pore to delineate the impact of invisible B branches on the retention of copolymers. When the A backbone is in SEC mode, the presence of B branches near the critical condition leads to an increase in the attraction between the polymer and the pore surfaces; the entire graft copolymer is thus pulled into the pore and K(A-g-B) > K(A). When the A backbone is in LAC mode, connecting a B branch to the A backbone increases the entropic repulsion between the polymer and the pore surfaces; the entire graft copolymer is thus pushed away from these surfaces and K(A-g-B) < K(A). While chain connectivity is sufficient to result in the invisible B branches influencing the elution of A-g-B copolymers, simulations using SAW chains show that the excluded volume effect increases this influence. In particular, SAW chains can experience a significant reduction in chain entropy if B branches are attached to the A backbone at graft points very near chain surfaces due to the excluded volume effect.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data related to this article can be found at http://dx.doi.org/10.1016/j.polymer.2015.04.067.

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